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B.M.S. COLLEGE OF ENGINEERING, BENGALURU - 19

(Autonomous Institute, Affiliated to VTU)

DEPARTMENT OF MACHINE LEARNING

Continuous Internal Evaluation (CIE) - 1

Course Code	:	24AM7PCTSA	Course Title	:	Time Series Analysis		
Semester	:	7	Maximum Marks	:	40	Date	: 25 Sep 2025
Faculty Handling the Course	:	Dr. GS, Prof. MK, Prof. SR					

Instructions: *Internal Choice is provided in Part - C*

Part - A

Total 5 Marks (No Choice)

Q. No.	Question	Marks	CO	Level
1a	How is time series analysis different from other Machine Learning methods on static data? Highlight its unique characteristics.	5M	1	1

Part – B

Total 15 Marks (No Choice)

Q. No.	Question	Marks	CO	Level
2a	While analyzing user adoption of a new AI-powered application, a company obtained the least squares trend equation: $y = 1600 + 200x$ (Origin is 1980) where x represents years, y is the annual number of active users. The platform can support a maximum of 3600 users at a time, and it is assumed that the trend continues at least for the next decade. i. What is the average annual increase in the number of users? ii. By which year will the expected active user base equal the system's maximum capacity. iii. Estimate the user base in 1995. How much does this exceed the platform's physical capacity?	5M	1	2
2b	Calculate the seasonal indices using the simple averages method for a monthly dataset. Write the steps in brief.	5M	1	2
2c	Analyze the various components of time series data. Explain how they can be modelled.	5M	1	2

Part – C

Total 20 Marks (Internal Choice)

Q. No.	Question	Marks	CO	Level
3a	A company tracks the monthly active users (in thousands) of its AI chatbot from 1990 to 1999 (data below). Using a four-period moving average under	10M	1	3

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a multiplicative model, compute the trend values and obtain the trend-eliminated (detrended) series

Year	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999
Users(k)	75	60	55	60	65	70	70	75	85	100

Highlight the main disadvantage of trend estimation by moving average method by observing how many trend values are actually obtained.

OR

3b	The dataset below shows the accuracy scores (%) of a machine learning model recorded annually during deployment (1992–1997)	**10M**	1	3

Year	1992	1993	1994	1995	1996	1997
Accuracy(%)	56	55	51	47	42	38

i. Compute the trend values using the semi-averages method.
ii. Compute the trend values using the least squares method.
iii. Plot the original accuracy data along with the trends obtained from both methods on the same graph.
iv. Comment on the improvement of the least squares method over semi-averages by calculating the error sum of squares.
(Where error = difference between actual accuracy and trend values).

4a	Calculate the seasonal indices by the 'ratio to moving average' method from the following data.	**10M**	1	3

Year	Quarters			
	I	II	III	IV
2005	75	60	54	59
2006	86	65	63	80
2007	90	72	66	85
2008	100	78	72	93

OR

4b	Obtain the seasonal indices by the link relative method, for the following data:	**10M**	1	3

Quarter	Years			
	1996	1997	1998	1999
I	30	35	31	31
II	26	28	29	31
III	22	22	28	25
IV	31	36	32	35

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B.M.S. COLLEGE OF ENGINEERING, BENGALURU - 19
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DEPARTMENT OF MACHINE LEARNING

Continuous Internal Evaluation (CIE) - 1

Course Code	:	24AM7PCLLM	Course Title	:	Natural Language Processing for LLM		
Semester	:	7	Maximum Marks	:	40	Date	: 26-09-2025
Faculty Handling the Course	:	Dr. MP, Dr. SK, Dr. SBS					

Instructions: *Internal Choice is provided in Part - C*

Part - A

Total 5 Marks (No Choice)

Q. No.	Question	Marks	CO	Level
1a	Outline different phases of Natural Language Processing (NLP) with a neat diagram by giving suitable example at each phase.	5M		1

Part – B

Total 15 Marks (No Choice)

Q. No.	Question	Marks	CO	Level
2a	Justify the role of regular expressions in NLP. Provide regular expressions for the following: i. Email address ii. Date in the format DD/MM/YYYY with a year between 1900 and 2099.	5M	1	3
2b	Calculate the tf-idf score for the following words by considering the given term frequency and total documents as 37.	5M	1	3
2c	Analyze the need of text normalization? Classify the behaviors of lemmatization and stemming with examples for each.	5M	1	2

Word	df
Romeo	1
Salad	2
Forest	12
Battle	21
Fool	36
Good	37
sweet	37
Juliet	4

explain tf-idf

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Part – C

Total 20 Marks (Internal Choice)

Q. No.	Question	Marks	CO	Level
3a	Identify the type of ambiguity and all possible meanings arise from each of the sentences given. i. Time flies like an arrow. ii. Visiting relatives can be boring. iii. Are you still watching? iv. The teacher asked the student to bring his book to class. v. She observed the painting with a sharp eye.	10M	1	3
	OR			
3b	Apply Byte Pair Encoding for the following words with their frequencies. Write algorithm for the same. ("hug", 10), ("pug", 5), ("pun", 12), ("bun", 4), ("hugs", 5).	10M	1	3
4a	Given raw co-occurrence count vectors (contexts: water, finance, nature) for three words: • river=[300, 5, 200] • bank=[50, 400, 10] • money=[5, 450, 2] i. Compute the cosine similarity for the pairs: A. cos(river,bank) B. cos(river,money) C. cos(bank,money) ii. Convert the cosine values for pairs (A,B,C) to angles in degrees and determine the word that is closer in meaning to money.	10M	2	3
	OR			
4b	Given co-occurrence counts from a corpus: Total number of observed word–context pairs N=10,000 Marginal counts: • word counts: count(cat)=500, • count(dog)=800, • count(bank)=200 • context counts: count(pet)=900, • count(money)=700 Joint (co-occurrence) counts: • count(cat,pet)=300 • count(dog,pet)=600 • count(bank,money)=150 i. Compute the PMI (base-2) for each of these three pairs: PMI(cat, pet) PMI(dog, pet) PMI(bank, money) ii. Compute the positive PMI (PPMI) for each pair. iii. Identify which word–context association is strongest and why PMI is useful in this scenario.	10M	2	3

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B.M.S. COLLEGE OF ENGINEERING, BENGALURU - 19
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DEPARTMENT OF MACHINE LEARNING

Continuous Internal Evaluation (CIE) - 1

Course Code	:	24AM7PECYS	Course Title	:	Cyber Physical Systems		
Semester	:	7	Maximum Marks	:	40	Date	: 26 – Sep - 2025
Faculty Handling the Course	:	Prof. Supriya P					

Instructions: *Internal Choice is provided in Part - C*

Part - A
Total 5 Marks (No Choice)

Q. No.	Question	Marks	CO	Level
1a	Define Cyber-Physical Systems (CPS) and list the key components of CPS with a diagram.	5M	1	1

Part – B
Total 15 Marks (No Choice)

Q. No.	Question	Marks	CO	Level
2a	Elaborate on the design principles of Cyber-Physical Systems.	5M	1	2
2b	Explain the steps involved in Model-Based Design (MBD) with a diagram. List the frameworks commonly utilized in this methodology.	5M	1	2
2c	In the context of autonomous drones delivering packages in a smart city, which key safety invariants should be established to ensure their safe operation during these activities?	5M	1	2

Part – C
Total 20 Marks (Internal Choice)

Q. No.	Question	Marks	CO	Level
3a	i. Analyze synchronous reactive components in CPS by defining them and explaining their principles. ii. Design an event-triggered component SecondToMinute with the input event variable second and the output event variable minute such that minute is present every 60th time the event second is present.	10M	3	3

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	OR			
3b	i. Distinguish between Combinational Circuits and Sequential Circuits. ii. The combination of two components, where one waits for the other and disrupts a consistent order of updates, results in non-determinism. Justify. iii. Analyze and explain the different approaches for composing components in Cyber-Physical Systems (CPS).	10M	3	3
4a	Design a sensor-triggered combinational component called TemperatureAlert with two input variables, currentTemp and thresholdTemp, of type natural numbers (nat), and an input event variable named alertSignal. The output variable alert should be of type event(bool), such that the value of alert indicates true when currentTemp exceeds thresholdTemp during rounds in which the alertSignal is present.	10M	2	3
	OR			
4b	Provide pseudocode for an alarm system that integrates three components: MotionSensor, DoorSensor, and Alarm. The alarm should activate if the motion or door sensor detects an intrusion.	10M	2	3

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B.M.S. College of Engineering, Bengaluru-560019

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CIE 1(Internal test)-SET C

Programme: B.E. **Semester: VII**

Branch: Common to all Branches **Duration: 1 h**

Course Code: 25MA7HSIKL **Max Marks: 40**

Course: INDIAN KNOWLEDGE SYSTEMS

Instructions: 1. Answer all questions, by choosing correct option

Q. No	Question	Marks
1.	What does Indian Knowledge System (IKS) primarily emphasize? A) Rapid industrialization B) Materialism C) Holistic and interdisciplinary learning D) Western scientific principles	1
2.	Which texts form the foundation of Indian Knowledge Systems? A) Quran and Bible B) Tripitaka and Zend Avesta C) Vedas and Upanishads D) Manusmriti and Arthashastra	1
3.	Which Veda is considered the oldest? A) Yajurveda B) Samaveda C) Atharvaveda D) Rigveda	1
4.	What is the primary content of the Yajurveda? A) Hymns for gods B) Rituals and sacrificial ceremonies C) Healing techniques D) Musical chants	1
5.	Which Veda is known for its musical chants? A) Rigveda B) Atharvaveda C) Yajurveda D) Samaveda	1
6.	Which Veda contains knowledge related to medicine and daily life? A) Rigveda B) Yajurveda C) Samaveda D) Atharvaveda	1
7.	What does the term 'Shruti' refer to? A) Remembered literature B) Classical poetry C) Eternal texts heard by sages D) Commentaries on rituals	1
8.	What does 'Smriti' mean in the context of Vedic literature? A) Eternal truth B) Heard scriptures C) Remembered texts that are flexible D) Collections of hymns	1
9.	How many hymns are there in the Rigveda? A) 500 B) 1028 C) 100 D) 3000	1
10.	What is the correct sequence of the four layers of each Veda? A) Upanishads, Aranyakas, Samhitas, Brahmanas B) Samhitas, Brahmanas, Aranyakas, Upanishads	1

	C) Brahmanas, Samhitas, Upanishads, Aranyakas D) Aranyakas, Samhitas, Brahmanas, Upanishads	
11.	Which of the following is not true about coexistence approach to indigenous knowledge (IK)? A) It devalues Western or Indigenous resource management practices and the knowledge that informs them B) It does not devalue Western or Indigenous resource management practices and the knowledge that informs them C) It does not allow for the domination of one over the other D) It values both systems	1
12.	Which of the following best characterizes traditional knowledge? A) It is represented by a villager with a rich experiential traditional knowledge heritage B) It is represented by a villager with a rich experiential modern knowledge heritage C) It is represented by a scientist with a rich experiential traditional knowledge heritage D) It is represented by a scientist with a rich experiential modern knowledge heritage	1
13.	Which of the following is an example of Traditional Ecological Knowledge (TEK)? A) Using CRISPR gene editing for disease resistance B) Conducting clinical drug trials for new medicines C) Practicing controlled burning to regenerate forests D) Designing artificial intelligence algorithms	1
14.	How does Western Knowledge (WK) primarily differ from Traditional Knowledge? A) WK is holistic and spirituality-based, while TK is data-driven B) WK relies on empirical evidence and experiments, while TK is based on community experiences C) WK is only used in the medical field, while TK is used in all aspects of life D) WK is always superior to TK in solving environmental issues	1
15.	Indigenous knowledge is regarded as inseparable and embodied in their affinities with ? A) Science B) Technology C) Cosmos D) Land	1
16.	What is one of the strengths of Western Knowledge? A) It is always in harmony with nature B) It is based on systematic learning and continuous advancement C) It only focuses on spiritual and social aspects D) It is entirely oral and undocumented	1
17.	What is the role of Traditional Knowledge in biodiversity conservation? A) It promotes industrial expansion over forest protection B) It helps in sustainable resource use and ecosystem balance C) It discourages conservation efforts in favor of urbanization D) It has no impact on biodiversity conservation	1
18.	Which of the following best describes the Scientific Method in Western Knowledge? A) Observation → Questioning → Hypothesis → Experimentation → Analysis → Peer Review B) Guesswork → Belief → Tradition → Conclusion C) Oral Transmission → Spiritual Understanding → Validation D) Innovation → Storytelling → Rituals	1
19.	What is a significant contribution of Traditional Knowledge to modern medicine? A) The discovery of antibiotics from synthetic chemicals B) The development of	1

	vaccines through genetic engineering C) The use of medicinal plants, such as quinine for malaria treatment D) The production of painkillers using petroleum-based ingredients	
20.	How can governments help in preserving Traditional Knowledge? A) By disregarding Indigenous practices in favor of Western education B) By providing legal recognition and intellectual property rights to Indigenous communities C) By preventing Indigenous groups from passing down their knowledge D) By discouraging the use of Traditional Knowledge in any scientific research	1
21.	According to Indian Philosophy merit and demerit are considered as A) spshta B) adrsta C) drsya D) none of these	1
22.	Rigveda is the oldest of all vedas. A) True B) False	1
23.	Agni , Indra and Varuna are the gods chanted in Rigveda. A) False B) True	1
24.	Which veda deals with rituals, sacrificial rites, and laws of governance A) Samaveda B) Yajurveda C) Rigveda D) Arthaveda	1
25.	Samaveda is the basis for A) Medicine B) ethics C) mental health D) Indian Classical Music.	1
26.	Which veda deals with medicine, governance, ethics, and mental health. A) Samaveda B) Yajurveda C) Rigveda D) Arthaveda	1
27.	The Auxiliary Sciences of Kalpa describes A) social duties and rites B) atymology of difficult vedic word C) basis of Sanskrit language D) proper sound pronunciation and intonation	1
28.	Upangas that introduce practical applications to the vedic corpus are A) Dharma Shastras and Arthashastra B) Itihasa-Puranas C) Nyaya and Mimamsa D) All of the above	1
29.	Traditionally there are Upanishads. A) 108 B) 110 C) 100 D) 101	1
30.	In the Mandukya Upanishad, AUM is described as an A) image of wakefulness B) dreams C) deep sleep D) All of the above	1
31.	Which layer of the Vedas is intended for hermits and forest dwellers? A) Samhitas B) Brahmanas C) Aranyakas D) Upanishads	1
32.	Indian Philosophical system can be classified in to systems A) Vedic B)Non Vedic C) Both a and b D) None of the above	1
33.	---------- is considered as the foundation of Indian Philosophy A) Samhitas B) Brahmans C) Aranyakas D) Upanishads	1

34.	Philosophical knowledge does not aim to satisfying our theoretical interest, but also to A) realizing God B) realizing perfect good in life C) realizing the highest truth in life. D) None of these	1
35.	Indian philosophy is essentially ----------in nature. A) materialistic B) spiritualistic C) optimistic D) ritualistic	1
36.	Indian Philosophy is called Darsana, which means A) reality B) truth C) system D) vision	1
37.	According to Indian Philosophy God is the creator, the preserver and the ---------- ---- of the cosmos A) successor B) destroyer C) master D) none of these	1
38.	All schools of Indian philosophy except the ----------believe in the Law of Karma. A) Buddhism B) Jainism C) Carvaka D) Samkhya	1
39.	As we sow, so we reap is related with A)Law of Reality B) Law of Causation C) Law of liberation D)Law of Karma	1
40.	Punya is a result of : A) performance of a duty B) violation of duty C) rejection of duty D) none of these	1

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